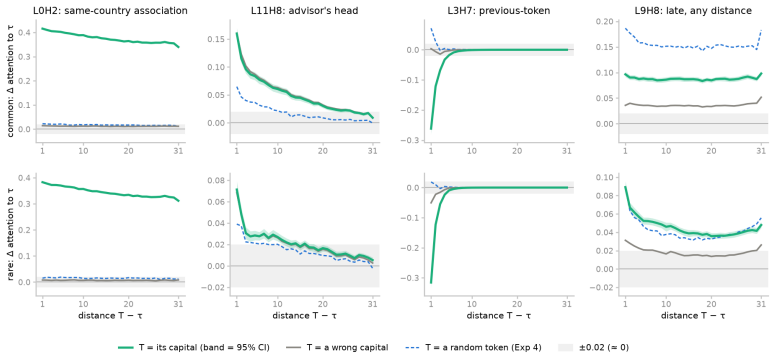
Where the “no-op” lives: three results

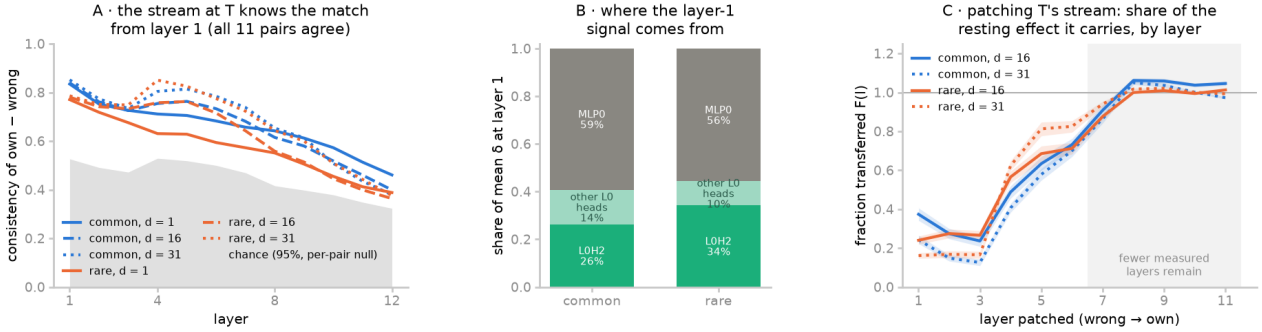
GPT-2 small. Random-token rows with a country at position τ and its capital (or a wrong capital) as the final token T. Each pair of sequences differs at one position only. 200 / 100 rows $\times$ 11 country–capital pairs per filler type. Every analysis was pre-registered and independently re-checked.

1 The matching capital lowers late resting at every distance, through specific head families



- Late-layer attention parked on BOS drops by **0.02–0.04** with the country’s own capital at T, against a wrong capital, at every distance from 1 to 31 tokens. A placebo word at τ gives ≈ 0.
- **Same-country heads from layer 0.** L0H2 gives the country 0.43 of its attention with its own capital and 0.03 with a wrong one. L4H10, L3H0 and L1H11 do the same, but for any same-country pair (demonyms, cities), not specifically “capital of”.
- The advisor’s L11H8 is not a previous-token head: it puts only 6–8% on the previous token. Only L4H11 is a pure previous-token head.

2 In the residual stream, the match enters at layer 1 and the resting effect is assembled in layers 3–7



- **A.** From layer 1, the stream at T carries an own-vs-wrong direction shared by all 11 pairs (exact per-pair null $p = 1/1024$ at every layer, margin 1.3–1.6$\times$).
- **B.** At layer 1 it comes from L0H2 (26–34%), the other layer-0 heads (10–14%) and the first MLP (56–59%). It is a “matching country attended” signal, not the country’s identity: a mismatched country looks like filler to T.
- **C.** Patching: T’s stream carries 12–37% of the resting effect by layer 3, a step at layer 4, and 86–94% by layer 7. From layer 8 it reproduces the effect within $\pm 6\%$. Context still matters at layers 6–7.

3 In the writes, “no-op” means a constant offset

Heads that rest on BOS write less varied vectors ($\rho = -0.67$ / -0.70 across 144 heads). But the BOS term is effectively a fixed vector, about 2% of write variance, and per unit of attention *not* on BOS resting heads write as much variety as active heads or more. So the no-op is attention reweighted onto a constant, not quieter reading.

Did not hold: distance dependence falling with depth (it rises); clean topological clusters of heads; the four same-country heads carrying $\geq$20% of the patching effect (18–36%, fragile).

The main caveat. Everything rests on the same 11 country–capital pairs. In paired tests the effective n is 11, so “beyond the null” means consistent across rows, not general.

To decide together

1. Generalise first (more countries; cities and demonyms at τ to separate “capital of” from same-country relatedness), or go deeper (the first MLP’s 56–59%; the layer 3–7 assembly)?
2. Is “no-op = attention reweighted onto a constant BOS vector” the right claim to build on?